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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/265,212	06/02/2023	Jonne STEEMAN	D5481.10120US01	6511

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EXAMINER	
CARLSON, MARC	

ART UNIT	PAPER NUMBER
3723	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/265,212	Applicant(s) STEEMAN, Jonne	
	Examiner MARC CARLSON	Art Unit 3723	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 6/5/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-15 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-15 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 6/2/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 6/2/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Claim Interpretation

The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

1. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

2. This application includes one or more claim limitations that use the word "means", "step", or a generic placeholder but are nonetheless not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph because the claim limitation(s) recite(s) sufficient structure, materials, or acts to entirely perform the recited function. Such claim limitation(s) is/are: "core element" in Claims 1, 3-7, 9, 12, and 13, "cleaning elements" in Claims 1-3 and 8-12, , and "elements" in Claim 3.

Because this/these claim limitation(s) is/are **not** being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are **not** being interpreted to cover only the corresponding structure, material, or acts described in the specification as performing the claimed function, and equivalents thereof.

If applicant intends to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to remove the structure, materials, or acts that performs the claimed function; or (2) present a sufficient showing that the claim limitation(s) does/do not recite sufficient structure, materials, or acts to perform the claimed function.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

3. Claims 1-7, 9, and 11-15 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Udy et al. US 2018/0070785 (hereafter Udy et al.).

a. Regarding Claim 1, Udy et al. anticipates:

1. An agitator (agitator 18) configured:

to be used in a suction head (cleaning head 12) that is configured to be applied in a vacuum cleaner (surface cleaning apparatus 10) and to perform a cleaning action on a surface when the suction head is in an operational cleaning position relative to the surface (shown in Figure 2); and

to be rotatably arranged in the suction head and includes a core element (elongated agitator body 44) and flexible cleaning elements (bristles 40) arranged on the

core element (in continuous rows 46a-n or replaced by a continuous rubber blade and/or felt, Paragraph [0065]),

wherein an outer portion of each of the cleaning elements is configured to be moved over the surface during a part of each revolution of the agitator when the agitator is arranged in the suction head and the suction head is in the operational cleaning position (Paragraph [0066]),

wherein the cleaning elements are arranged along the length of the agitator in a cleaning field (hair migration portions 41 – two shown in Figure 3),

wherein the cleaning field is interrupted by at least one air passage (collection area 42) that is configured to enable an airflow between internal edges of the cleaning field delimiting the at least one air passage (best shown in Figure 3),

wherein outside of the at least one air passage, the cleaning elements are arranged at close range relative to each other (continuous rows 46a-n or replaced by a continuous rubber blade and/or felt, Paragraph [0065]) whereby an airflow between the cleaning elements is disabled or at most enabled to a significantly smaller extent than at the position of the at least one air passage (shown in Figure 3), and

wherein the cleaning field (hair migration portions 41 – two shown in Figure 3) spans a major part of the core element (elongated agitator body 44) in a peripheral direction around the core element (as shown in Figure 3, two hair migration portions 41 cover most of the length (except collection area 42) and the hair migration portions include the entire circumference of the elongated agitator body 44 within the areas labeled in Figure 3).

b. Regarding Claim 2, Udy et al. anticipates:

2. The agitator according to claim 1, wherein either the at least one air passage (collection area 42) is free from the cleaning elements (bristles 40)(Figure 3) or the at

least one air passage does include the cleaning elements but at a significantly lower packing density than the cleaning field (not selected).

c. Regarding Claim 3, Udy et al. anticipates:

3. The agitator according to claim 1, wherein the at least one air passage (collection area 42) includes elements (region 60 with a convex shape, Paragraph [0050], and Figure 16) extending from the core element (elongated agitator body 44) along a shorter radial distance than the cleaning elements (bristles 40).

d. Regarding Claim 4, Udy et al. anticipates:

4. The agitator according to claim 1, wherein the at least one air passage (collection area 42) extends in a peripheral direction around the core element (elongated agitator body 44)(shown in Figure 3).

e. Regarding Claim 5, Udy et al. anticipates:

5. The agitator according to claim 4, wherein the at least one air passage (collection area 42) extends perpendicular to a longitudinal axis of the core element (elongated agitator body 44)(shown in Figure 3).

f. Regarding Claim 6, Udy et al. anticipates:

6. The agitator according to claim 4, wherein the at least one air passage (collection area 42) spans an entire periphery of the core element (elongated agitator body 44)(shown in Figure 3).

g. Regarding Claim 7, Udy et al. anticipates:

7. The agitator according to claim 6, wherein the cleaning field (hair migration portions 41 – two shown in Figure 3) spans the entire periphery of the core element (elongated agitator body 44)(shown in Figure 3).

h. Regarding Claim 9, Udy et al. anticipates:

9. The agitator according to claim 1, wherein the cleaning elements (bristles 40) are arranged on the core element (elongated agitator body 44) in tufts (tuft forming continuous rows 46a-n, Figure 7).

i. Regarding Claim 11, Udy et al. anticipates:

11. The agitator according to claim 1, wherein the agitator (agitator 18) comprises at least one agitator element (continuous rubber blade and/or felt, Paragraph [0065]) in addition to the cleaning elements (bristles 40)(Paragraph [0065] - In at least one embodiment, the rubber blade may extend along adjacent to one or more rows 46a-n of bristles as described herein), wherein the at least one agitator element is significantly less flexible than the cleaning elements (continuous rubber blade would be expected to be stiffer than bristles), and wherein an outer portion of the at least one agitator element is configured to be moved over the surface during a part of each revolution of the agitator when the agitator is arranged in the suction head and the suction head is in the operational cleaning position (Figure 2 and Paragraph [0065).

j. Regarding Claim 12, Udy et al. anticipates:

12. The agitator according to claim 11, wherein the at least one agitator element (continuous rubber blade and/or felt, Paragraph [0065]) is arranged on at least one strip-shaped body, and wherein the at least one strip-shaped body is arranged to spiral

around the core element at a position of outer portions of the cleaning elements (shown in Figure 3).

k. Regarding Claim 13, Udy et al. anticipates:

13. A suction head (cleaning head 12) configured to be applied in a vacuum cleaner (surface cleaning apparatus 10) and to perform a cleaning action on a surface when the suction head is in an operational cleaning position relative to the surface (Figure 2), the suction head comprising:

a housing (head body 13) including a coupling area (duct 36) configured to enable coupling of the housing to an air suction source (vacuum source 32) of the vacuum cleaner, and

the agitator (agitator 18) according to claim 1 (see Claim 1 rejection above),

wherein the agitator is arranged in the housing and the housing is configured to expose a portion of the agitator to the surface (shown in Figure 2),

wherein the agitator is rotatable in the housing about a rotation axis (R) extending in a direction of a longitudinal axis of the core element (shown in Figure 2),

wherein the agitator is configured to enable an airflow passing the agitator from a position on the surface upstream of the agitator in a direction towards the coupling area of the housing at a position of the at least one air passage (collection area 42) in the cleaning field (hair migration portions 41 – two shown in Figure 3) of the agitator when the suction head is in the operational cleaning position (shown in Figure 2), and

wherein the agitator is configured to disable or at most enable to a significantly smaller extent an airflow passing the agitator from the position on the surface upstream of the agitator in the direction towards the coupling area of the housing at the position

that is outside of the at least one air passage when the suction head is in the operational cleaning position (shown in Figures 2 and 3).

l. Regarding Claim 14, Udy et al. anticipates:

14. The vacuum cleaner, comprising an air suction source (vacuum source 32) and the suction head (cleaning head 12) according to claim 13.

m. Regarding Claim 15, Udy et al. anticipates:

15. The vacuum cleaner, comprising an air suction source (vacuum source 32) and the agitator (agitator 18) according to claim 1.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

> Determining the scope and contents of the prior art.
> Ascertaining the differences between the prior art and the claims at issue.
> Resolving the level of ordinary skill in the pertinent art.
> Considering objective evidence present in the application indicating obviousness or nonobviousness.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

4. Claims 8 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over Udy et al. US 2018/0070785 (hereafter Udy et al.) in view of design choice or optimal range obviousness.

a. Regarding Claim 8, Udy et al. teaches:

8. The agitator according to claim 1, wherein the cleaning elements (bristles 40) comprise fiber hairs, and wherein a linear mass density of the cleaning elements is lower than 300 g per 10 km.

Udy et al. discloses the use of bristles 40 that should be stiff enough to clean carpets without undesirable bending yet flexible enough to allow side to side bending as presented in Paragraph [0053]. Although mass density is an inherent property of the cleaning elements, Udy et al. does not disclose a value for the fibers that are employed in his device, therefore it may be possible that the fibers used by Udy et al. actually meet the claim limitation. It would have been obvious to one having ordinary skill before the effective filing date of the claimed invention to select a fiber with a mass density lower than 300 g per 10km

with the motivation to employ bristles that agitate debris from carpet, like the Applicant's device, since it has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416. See also *Ballas Liquidating Co. v. Allied industries of Kansas, Inc.* (DC Kans) 205 USPQ 331.

b. Regarding Claim 10, Udy et al. teaches:

10. The agitator according to claim 9, wherein a packing density of the cleaning elements (bristles 40) in the cleaning field (hair migration portions 41 – two shown in Figure 3) outside of the at least one air passage (collection area 42) is at least 15,000 cleaning elements per area of 1 cm by 1 cm (see discussion below).

Udy et al. discloses the use of bristles 40 that should be stiff enough to clean carpets without undesirable bending yet flexible enough to allow side to side bending as presented in Paragraph [0053]. In Figure 7 Udy et al. graphically shows a packing density of the bristles, however, Udy et al. does not provide a value for packing density employed in his device, therefore it may be possible that the fibers used by Udy et al. actually meet the claim limitation. It would have been obvious to one having ordinary skill before the effective filing date of the claimed invention that the bristle performance is directly related to the combination of fiber diameter, fiber length, packed bristle diameter, and packing density. Therefore, it would have been obvious through routine experimentation to adjust these values to achieve a desired stiffness with side to side flexibility.

Therefore, it would have been obvious that with a packing density of at least 15,000 cleaning elements per area of 1 cm by 1 cm, one could adjust the fiber diameter, fiber length, and/or packed bristle diameter to achieve a desired result, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. *In re Aller*, 105 USPQ 233.

Conclusion

5. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure can be found in form PTO-892 Notice of References Cited. Specifically, the prior art references include pertinent disclosures of vacuum cleaners with agitators including gaps in the bristle pattern.

6. Any inquiry concerning this communication or earlier communications from the examiner should be directed to MARC CARLSON whose telephone number is (571)272-9963. The examiner can normally be reached Monday-Thursday 6:30am-3:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, BRIAN KELLER can be reached on (571) 272-8548. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/MARC CARLSON/

Primary Examiner, Art Unit 3723